Scrubgnome-Face

Twig's photo edition. Strip the snitchy metadata out of your photos before you post them.

> "Great gears! That photo is blabbing where you stood, when, and with what. Hand it over." — Twig

Scrubgnome-Face is the photo member of the Scrubgnome family, alongside desktop Scrubgnome and the video app Scrubgnome Away from Home. It does one job well:

1. **Pick** photos (or *Share* them to "Scrub with Twig" from your gallery). One or a whole batch.
2. **See** what each one is leaking, in plain words: where, when, what phone, who, and hidden extras.
3. **Scrub & save** clean copies to `Pictures/Scrubgnome/` with random filenames.
4. **Share / upload** the clean copies.

Two flavours, same brain: - **Android app**: install the APK. - **Web version for iPhone / any browser** (`web/`): open the page in Safari, tap *Share* → *Add to Home Screen*, and it works like an app, even offline. It reads your photos locally and never uploads them; the page's security policy forbids it from sending data anywhere.

Privacy promise: the Android app has **no internet permission at all**. Everything happens on your phone.

What gets removed

Leak	Where it hides	What Twig does
GPS location	EXIF GPS block, XMP, IPTC city/country	removed
Date & time (and your time zone)	EXIF DateTime*, OffsetTime*, GPS date, XMP, PNG tIME	removed
Phone make, model, lens, software	EXIF, XMP, MakerNote vendor blob	removed
Who you are	Artist, owner name, camera/lens **serial numbers**, copyright, IPTC by-line	removed
Hidden preview thumbnail	EXIF IFD1 / JFXX (can show the **uncropped** original!)	removed
Edit history	XMP history (every edit, with app + timestamp)	removed
Motion-photo video	MP4 appended after the JPEG by Pixel/Samsung "motion photos"	cut off
HDR gain maps, extra images, C2PA records	MPF / APP11 / trailing data	removed
Comments & captions	JPEG/GIF comments, PNG text, IPTC captions	removed
The filename	`PXL_20261107_183522.jpg` *is* a timestamp	clean copy gets `photo_<random>.jpg`

Kept: the orientation tag, so photos don't turn sideways, plus the colour profile, so colours don't shift.

Lossless, every format: JPEG, PNG, WebP and GIF are rebuilt around their untouched compressed picture data. HEIC/HEIF (iPhone) and AVIF are cleaned **in place**: the metadata items are overwritten with an empty stand-in of the same size, so every pointer in the file stays valid. Not one pixel is re-encoded. After scrubbing, Twig re-scans the result and refuses to save it if anything survived.

Supported: JPEG, PNG, WebP, GIF, HEIC/HEIF, AVIF. Not supported: RAW/DNG, TIFF (use desktop Scrubgnome).

What it can't do: it removes *hidden* information only. A street sign, a face, a reflection or your house number in the picture itself is still there. (Face blurring is a maybe-someday idea.)

Install (no app store needed)

1. Open the repo's **Actions** tab → newest green **Build APK** run → download **ScrubgnomeFace-apk** (a zip holding the APK). Tagged versions also show up under **Releases**.
2. Copy the APK to your phone and tap it. Android will ask you to allow installs from that app (Files/Chrome). Allow it once.
3. Requires **Android 10 or newer**.
4. On first use Twig asks for **access to photo locations**. That lets him *show* you the real GPS (Android hides it from apps otherwise). Saying no is fine: he scrubs the GPS either way, he just can't show you the coordinates.

iPhone / web version

1. Open the web app's address in Safari (published from this repo via GitHub Pages once the repo is public; see `.github/workflows/pages.yml`).
2. Tap **Share → Add to Home Screen**. Twig gets his own icon and works offline after the first visit.
3. **Pick photos** → choose **Browse** to pick from Files (that hands over the untouched original). The Photo Library option may give Twig a converted copy, which is still fine to scrub.
4. **Scrub all** → **Save / share** → **Save Image** puts the clean copies in Photos.

The JavaScript scrubber produces byte-for-byte the same output as the Android (Java) one, and both match desktop Scrubgnome's Photos tab for JPEG, PNG, WebP and GIF.

Build it yourself

Easiest: push to GitHub and let Actions build it (see `.github/workflows/build-apk.yml`). Release signing uses the same four repo secrets as Away from Home (`SG_KEYSTORE_B64`, `SG_KEYSTORE_PASS`, `SG_KEY_ALIAS`, `SG_KEY_PASS`). Without them you get a debug APK.

Android Studio: *File → Open* this folder, wait for Gradle sync, press ▶ with your phone plugged in (USB debugging on).

Desktop self-tests (Java 17+, Node 22+, Python 3 with `pillow` + `pillow-heif`, and `exiftool`):

```
bash tools/selftest.sh        # Java core: scrub generated photos, verify with exiftool + pixel compare
bash tools/web-selftest.sh    # also: JS must match Java byte-for-byte
```

Project layout

```
app/src/main/java/com/kultrshok/scrubgnome/face/
├── core/ImageScrubber.java   ← the scrubber (pure Java, no Android; reusable module)
└── MainActivity.java         ← the one-screen UI (Twig's voice lives here)
web/
├── index.html                ← the web/iPhone app (one screen)
├── js/face.js                ← JS twin of ImageScrubber.java
├── js/app.js                 ← UI glue, Twig's voice
└── sw.js, manifest.webmanifest  ← offline + Add to Home Screen
tools/
├── make_samples.py           ← generates leaky test photos
├── SelfTest.java, web-selftest.mjs, verify.py, *.sh
```

No third-party libraries in the app. `core/ImageScrubber.java` and `web/js/face.js` drop straight into other projects.

License

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